Neutron Transport in Uranium Systems

A four-part study of uranium and water-moderated mixtures

Abstract

Monte Carlo neutron histories were simulated for pure U-238, unmoderated natural uranium, H2O with natural and 2%-enriched uranium, and D2O with natural uranium. The modeled medium is uniform and infinite. One million primary neutrons were tracked per case, with isotope and reaction selection weighted by atomic abundance and energy-dependent cross sections. For this study, K_∞ is the recorded thermal-fission daughter count divided by the initial neutron count. The unmoderated cases had $K_\infty = 0$. In the moderated cases, K_∞ reached unity between ratios 4-5 for natural U-H2O, 20-21 for 2%-enriched U-H2O, and 3-4 for natural U-D2O. H2O moderator captures increased strongly with water ratio, while D2O moderator captures remained below 1%.

Introduction

Neutron reaction balance depends on neutron energy, isotope composition, and moderator. This study compares unmoderated U-238 and natural uranium with H2O-moderated natural and 2%-enriched uranium and D2O-moderated natural uranium. The analysis focuses on collision-history trends, isotope-resolved fission, capture partitioning, and the study-defined K_∞ threshold across moderator ratios. Established Monte Carlo transport publications provide precedent for reporting model scope and nuclear-data choices [1, 2].

Computational method and scope

Collision histories are followed in a homogeneous, infinite mixture without spatial coordinates, leakage, or flight distances. At each collision, isotope and reaction probabilities are sampled from atom abundances and the available partial cross sections. The model uses pointwise KAERI NuChart ENDF/B-VII.1 data [3], with log-log interpolation between tabulated energies and endpoint-slope extrapolation above the tabulated range. Results are accumulated by isotope and reaction and reported for the specified 1–30 moderator-to-uranium ratios. Established transport-code publications emphasize documenting physics options and nuclear data [1, 2]; this study reports its model scope and tally definitions explicitly.

The analysis is descriptive: no replicate-based statistical uncertainties, convergence study, or benchmark comparison are reported. Conclusions therefore refer to trends within this implemented model. The benchmark literature provides a framework for a future matched validation study [4].

Notation

$K_\infty = N_g/N_0$, where N_g is the recorded set of daughters from fissions below 1 eV and N_0 is the initial neutron count. Unity is the criticality threshold used in this study.

A. Cross-Section Plots

The five isotope plots below provide the nuclear-data context for Parts B-E.

Energy-dependent cross sections

These isotope-specific cross sections determine the relative probabilities of elastic scattering, inelastic scattering, capture, and fission throughout the four simulations.

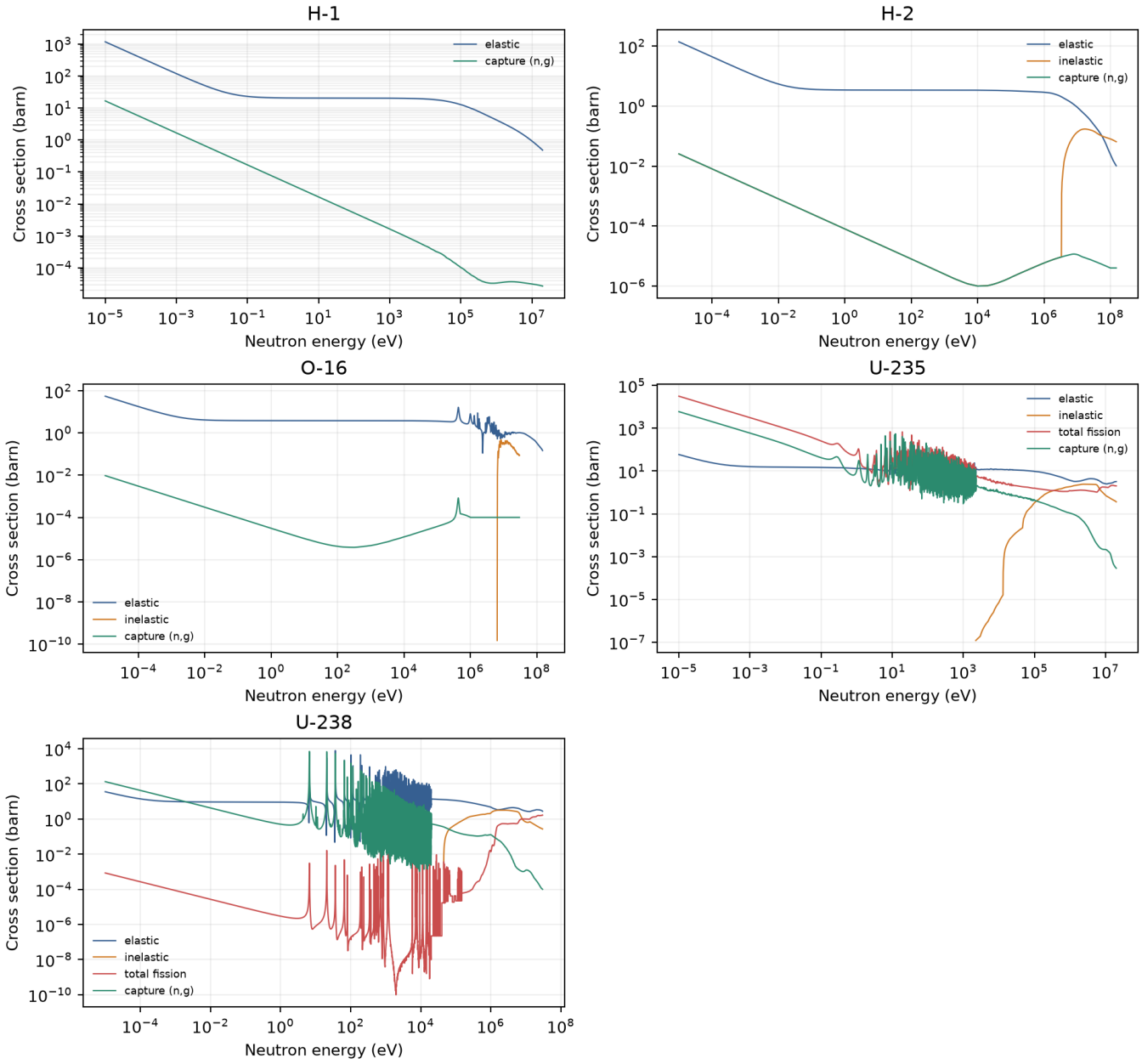


Figure 1. Cross sections for all modeled isotopes: H-1, H-2, O-16, U-235, and U-238. Both axes are logarithmic. Only reaction channels with supplied data are shown.

B. U-238 Only

The unmoderated U-238 case provides a reference for neutron slowing and fast-region fission in the absence of hydrogenous moderation.

1. Material and reaction results

The medium contains only U-238. 148,630 fission and 1,212,542 capture events were recorded across primary and daughter tracks. Fission occurred in the fast and resonance regions; no thermal fission occurred, giving $K_{\infty} = 0$. The energy histories show a sharp initial energy decrease followed by continued slowing over later collisions.

Composition	Tracks	Capture events	Fission events	Fission/capture	K∞
Pure U-238	1,361,172	1,212,542	148,630	0.123	0.000

2. Neutron source and energy history

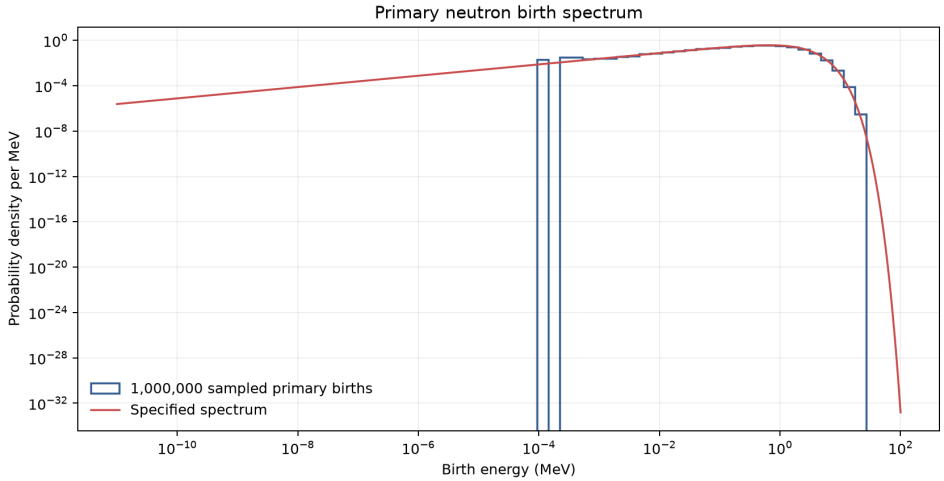


Figure 2. Sampled primary birth spectrum and its specified distribution.

3. Energy histories and terminal state

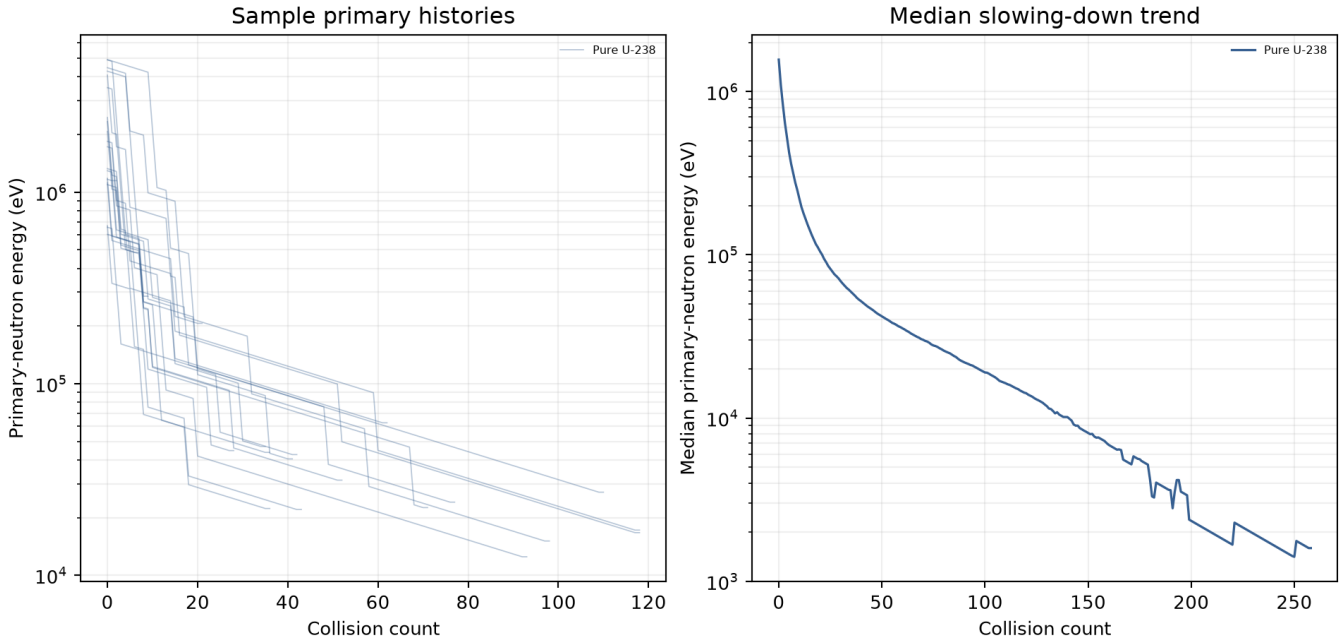


Figure 3. Sampled primary-neutron collision histories and median energy trend. Strong energy reduction occurs early in the track.

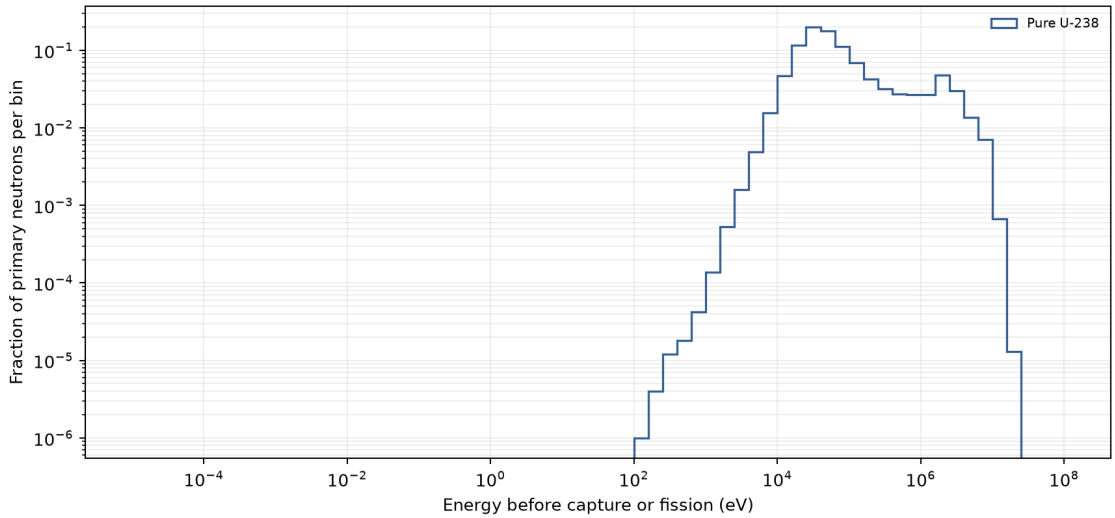


Figure 4. Primary-neutron energies immediately before capture or fission. All values belong to the initial population.



Figure 5. Number of collisions before primary-neutron termination; this is not a physical lifetime.

C. Natural Uranium

Unmoderated natural uranium provides an isotope-resolved comparison with the pure U-238 case.

1. Material and model

Natural uranium is represented by U-235:U-238 = 1:138. The recorded thermal-fission daughter count is zero ($K_\infty = 0$). U-235 accounts for 66,267 fissions, predominantly in the resonance range, while U-238 fission is predominantly fast in this run. Thus the minority U-235 isotope contributes a substantial resonance-fission component, whereas U-238 fission remains mainly fast.

2. Isotope-specific reaction balance

Uranium isotope	Elastic	Inelastic	Capture	Fission	Fission region
U-235	355,722	26,132	17,266	66,267	resonance
U-238	57,216,464	4,715,974	1,327,013	171,183	fast

Composition	Tracks	Capture events	Fission events	Fission/capture	K∞
Natural uranium	1,581,729	1,344,279	237,450	0.177	0.000

3. Energy histories

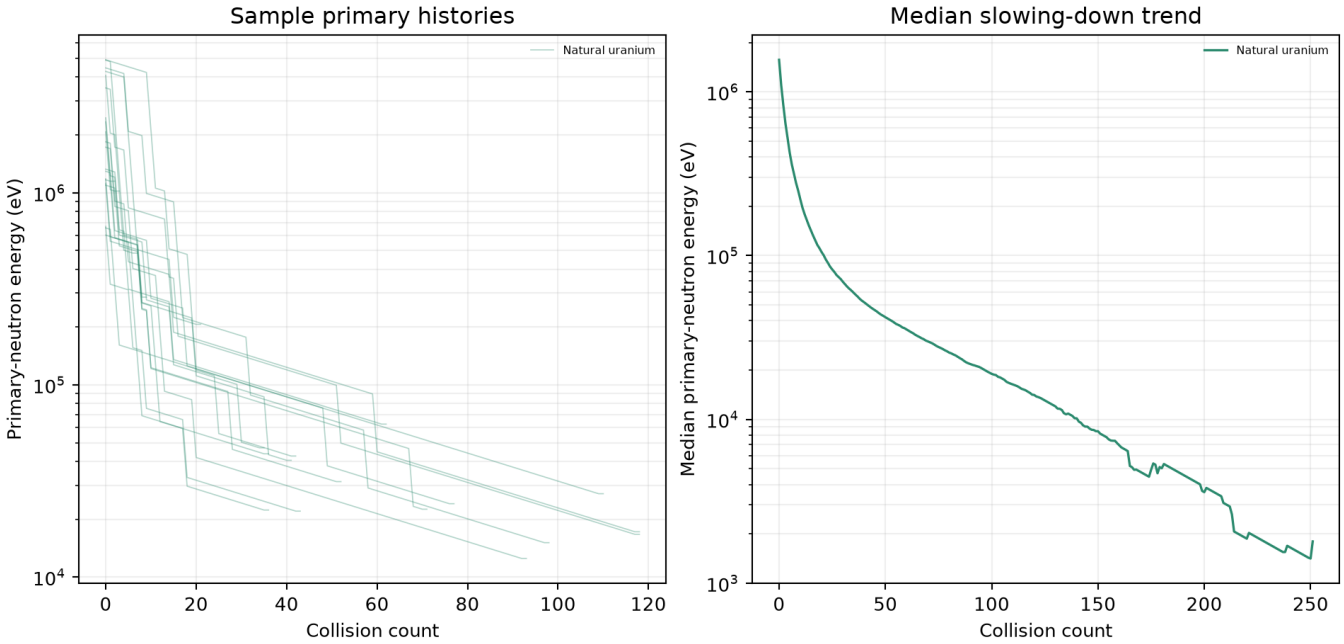


Figure 6. Sampled primary-neutron energy histories and median energy versus collision count in natural uranium.

4. Terminal distribution

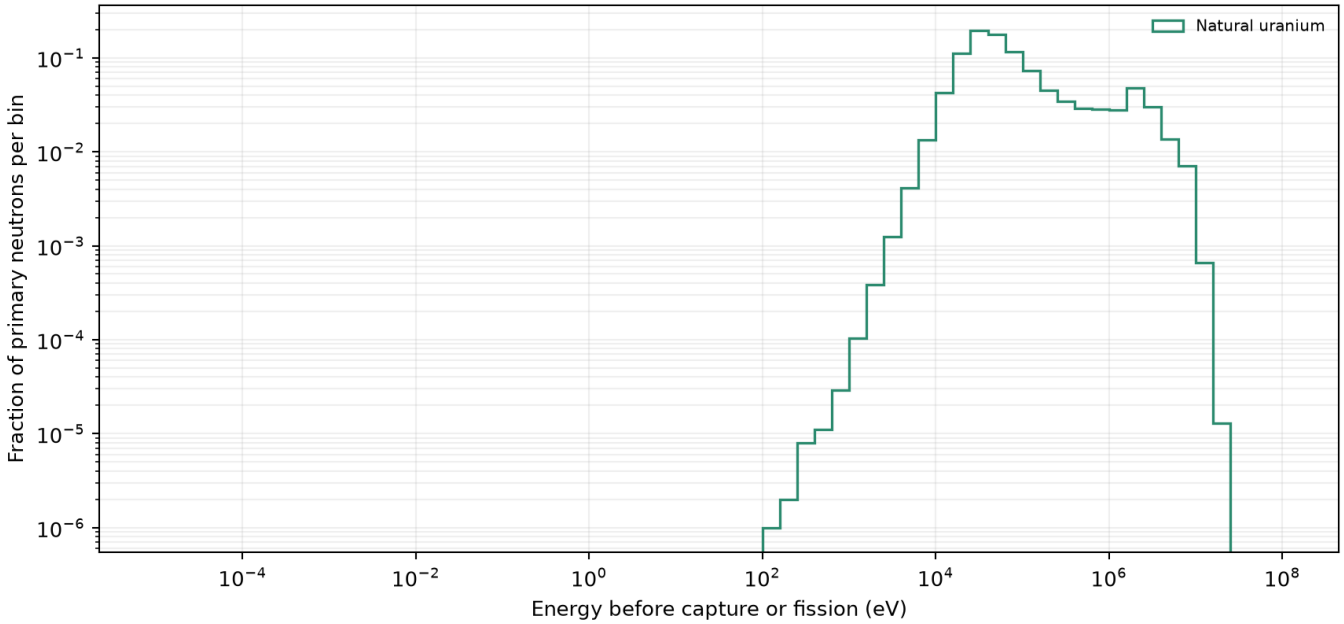


Figure 7. Terminal primary-neutron energies before capture or fission. The distributions retain substantial high-energy population because the modeled medium contains no moderator.

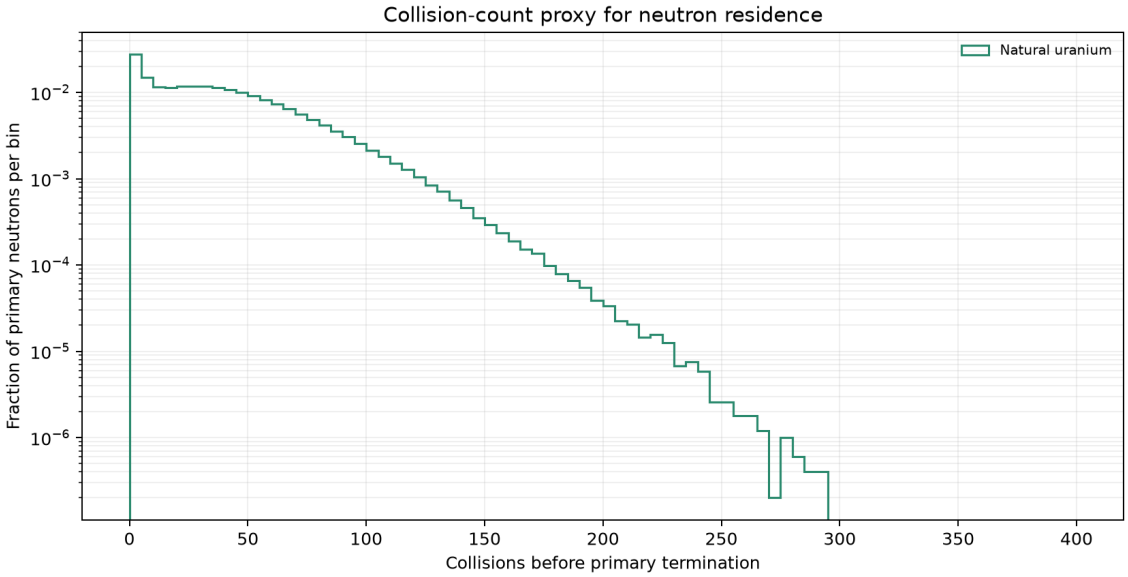


Figure 8. Primary-neutron collision counts before termination.

D. H2O-Moderated Uranium

The two light-water branches compare natural uranium (U-235:U-238 = 1:138) with nominally 2%-enriched uranium (1:49).

1. Material and model

Ordinary water was represented by two H-1 atoms and one O-16 atom per molecule. Natural uranium (U-235:U-238 = 1:138) and nominally 2%-enriched uranium (1:49) were each examined from 1 to 30 water molecules per uranium atom. K_{∞} is the recorded thermal-fission daughter count divided by the initial neutron count; the unity line marks the study's criticality threshold.

2. Moderator-ratio response

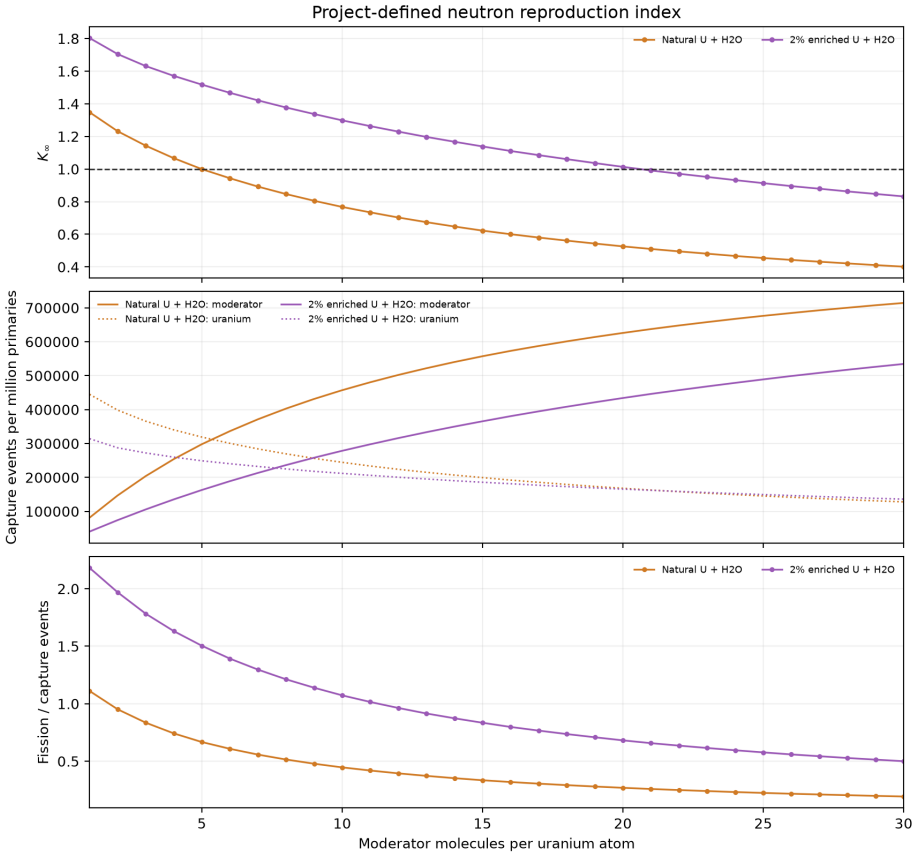


Figure 9. Project-defined K∞, moderator and uranium capture events per million primaries, and total fission-to-capture event ratio across both H2O branches.

3. Observed trends

Both K_{∞} curves decline as the H2O ratio increases, but the enriched branch remains higher throughout the sweep. Natural uranium falls from 1.3480 at ratio 1 to 0.4017 at ratio 30 and passes unity between ratios 4 and 5 (0.9994 at ratio 5). The 2%-enriched branch falls from 1.8027 to 0.8322 and passes unity between ratios 20 and 21 (0.9911 at ratio 21). The two branches therefore show substantially different moderator ranges around the unity threshold.

The capture curves show the reaction balance moving away from uranium and toward water as the moderator fraction increases. Moderator capture rises from 15.5% to 84.8% of all captures for natural uranium and from 11.4% to 79.7% for the enriched mixture. At the same time, the fission-to-capture event ratio decreases from 1.111 to 0.193 and from 2.181 to 0.500, respectively. Thus the changing K_{∞} response occurs alongside a strong shift in where neutron captures are recorded.

The isotope plots resolve the fission response behind these curves. From ratio 1 to 30, U-235 fissions per million primary neutrons decrease from 542,794 to 160,814 in natural uranium and from 731,542 to 333,296 in the enriched mixture. U-238 fissions decrease more sharply: 41,719 to 1,907 and 41,651 to 1,883, respectively. The higher U-235 fission counts in the enriched branch persist across the full range, while the reduction in U-238 fission is similar in both compositions.

4. Isotope-specific fission: natural uranium

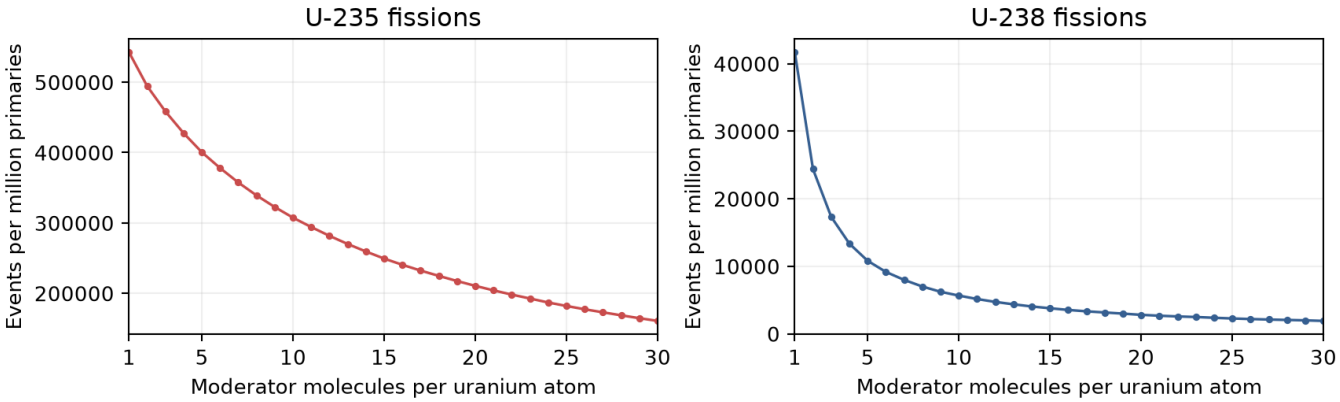


Figure 11. U-235 and U-238 fission events per million primary neutrons across the H2O ratio sweep. Both isotope-specific curves decrease, with the steeper decline in U-238.

In the natural-uranium series, U-235 fissions fall by about 70% between the lowest and highest water ratios. U-238 fissions fall by about 95%, from 41,719 to 1,907 events per million primaries. The curves separate the isotope contributions and show that the reduction is not uniform across the two uranium isotopes. U-235's share of the two isotope-specific fission counts rises from about 92.9% to 98.8% over the sweep. This change accompanies the decline in K_{∞} and the rising share of captures in the moderator.

5. Isotope-specific fission: 2%-enriched uranium

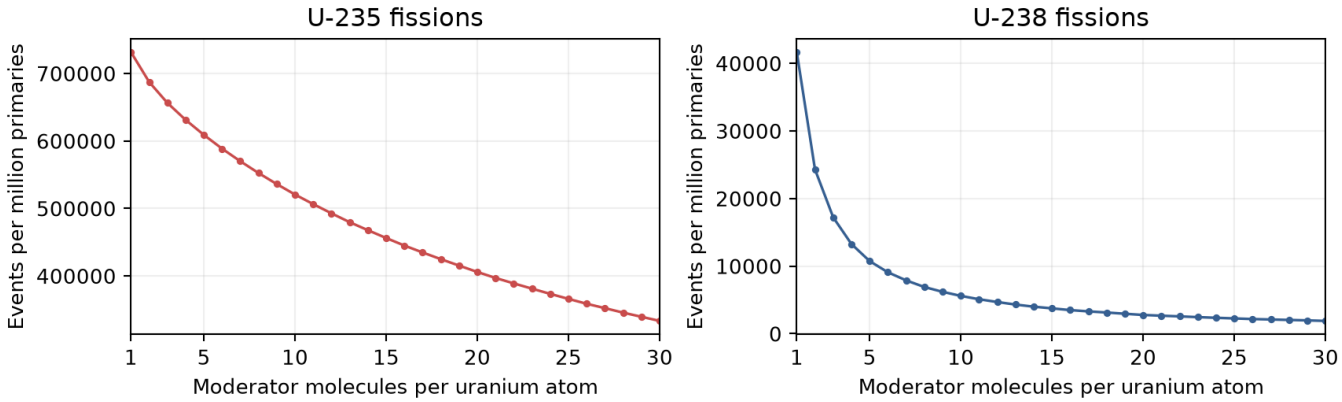


Figure 12. U-235 and U-238 fission events per million primary neutrons across the H2O ratio sweep. The enriched composition has higher U-235 fission counts throughout.

U-235 fissions decrease from 731,542 to 333,296 events per million primaries, a smaller proportional reduction than in the natural-uranium series. U-238 fissions fall from 41,651 to 1,883, closely matching the high-ratio endpoint of the natural uranium case. The persistent separation in U-235 counts reflects the distinct uranium compositions in these two sweeps, while both exhibit declining isotope-specific fission with increasing H2O ratio. U-235 accounts for about 94.6% of the two isotope fission counts at ratio 1 and 99.4% at ratio 30; the U-238 contribution becomes progressively smaller.

E. D2O-Moderated Natural Uranium

This part examines how the heavy-water moderator changes the reaction balance in natural uranium.

1. Material and model

Heavy water was represented by two H-2 atoms and one O-16 atom per molecule, mixed with natural uranium at U-235:U-238 = 1:138. The moderator ratio spans 1-30 D2O molecules per uranium atom. K_∞ is the recorded thermal-fission daughter count per initial neutron; unity is the criticality threshold used in this study.

2. Moderator-ratio response

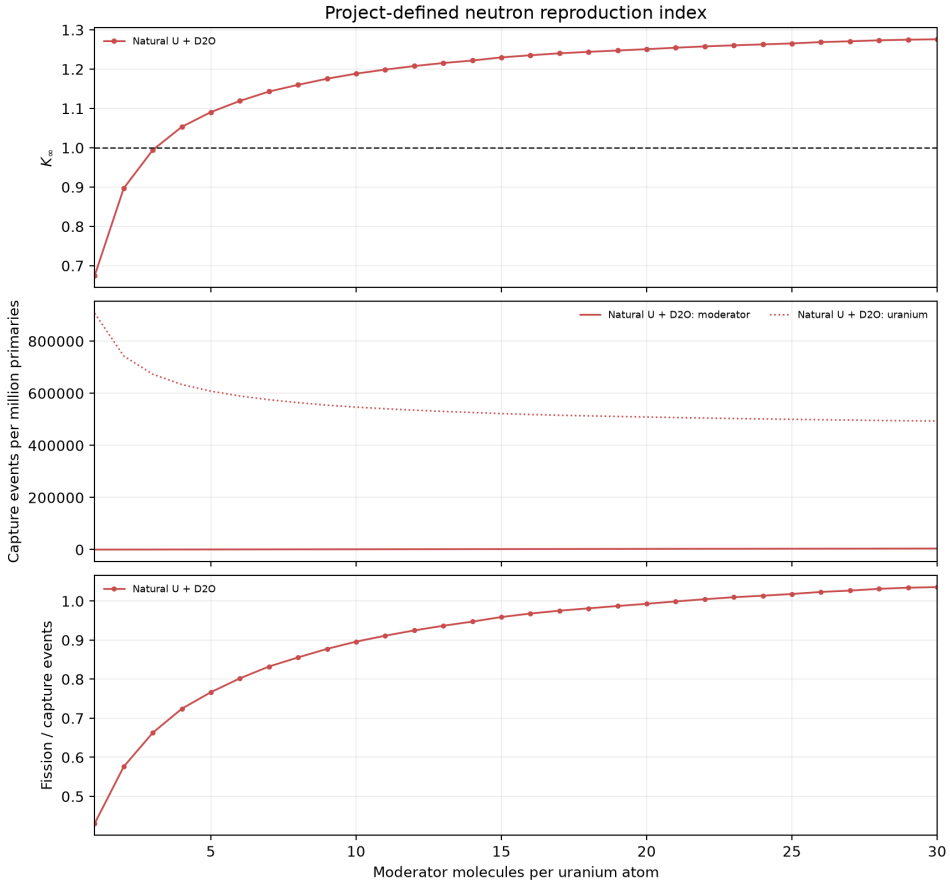


Figure 10. Project-defined K∞, moderator and uranium capture events per million primaries, and total fission-to-capture event ratio for natural uranium moderated by D2O.

3. Observed trends

Unlike the declining H2O curves, D2O K_∞ increases throughout this sweep, rising from 0.6746 at ratio 1 to 1.2759 at ratio 30. It passes unity between ratios 3 and 4; the closest tested sub-unity value is 0.9944 at ratio 3. This steady increase is accompanied by a rising fission-to-capture event ratio, from 0.431 to 1.036. The curves therefore show a progressive shift in the modeled reaction balance as D2O content increases.

The capture plot distinguishes D2O from H2O: moderator captures increase from 143 to 4,171 events per million primary neutrons, but remain a small share of total captures, rising from about 0.02% to 0.84%. Uranium captures decrease from 907,215 to 493,689 per million over the same interval. Most recorded captures therefore remain in uranium even at the largest D2O ratio, unlike the water-dominated capture balance at high H2O ratios.

4. Isotope-specific fission response

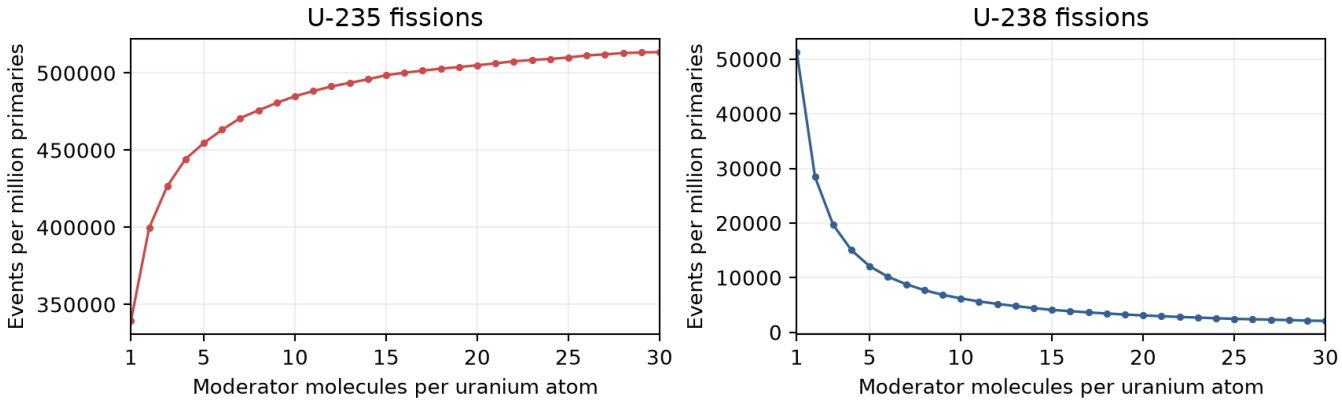


Figure 13. U-235 and U-238 fission events per million primary neutrons across the D2O ratio sweep.

U-235 fissions increase from 339,347 to 513,555 events per million primaries, while U-238 fissions decline from 51,298 to 2,070. U-235's share of the two isotope-specific fission counts therefore grows from about 86.9% to 99.6%. The opposing isotope trends accompany the rise in K_{∞} and in the overall fission-to-capture ratio, while uranium continues to account for most capture events. Together, the capture and fission plots show that increased D2O content changes both the balance of capture locations and the isotope distribution of fission.

F. Conclusion

Conclusion

The simulations show distinct moderation trends for the compositions examined. In the two H2O series, increasing the water-to-uranium ratio lowers the recorded K_{∞} and shifts an increasing share of captures from uranium to the moderator. The unity crossing occurs at ratios 4–5 for natural uranium and 20–21 for the 2%-enriched mixture. The isotope tallies show decreasing U-235 and U-238 fissions in both series, with U-238 fission decreasing most sharply.

The D2O natural-uranium sweep follows a different pattern: K_{∞} rises from 0.6746 to 1.2759 and crosses unity between ratios 3 and 4, while moderator capture remains below 1% of total captures. Across that sweep, U-235 fissions increase and U-238 fissions decrease. Taken together, these results identify moderator type and uranium enrichment as important variables in the reaction balances produced by this model.

These findings are bounded by the homogeneous, infinite collision-history model and its supplied reaction channels. The next steps for a stronger neutronics study are to quantify sampling uncertainty and convergence and to compare a suitably matched model with evaluated criticality benchmarks.

AI-use note

The code and report were generated with AI assistance and reviewed by a human at each stage.

All code is available at https://github.com/KedarSharma/Neutronics.